

Volatility Forecasting with Residual Learning

Eddie Liu | GARCH, LSTM, and empirical model evaluation

An additive LSTM reduces observed forecast MSE by **5.48%** versus GARCH with a fixed floor. But the uncertainty interval includes zero, and invalid raw variance forecasts remain a material limitation.

3,738

common test dates

178

walk-forward folds

2010-2024

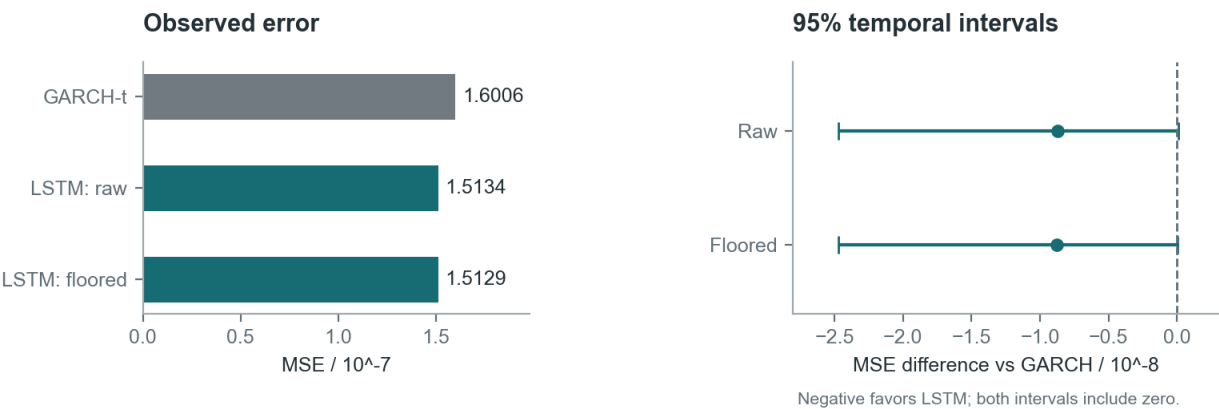
historical evaluation period

Can an LSTM learn GARCH's forecast errors?

The target is one-day S&P 500 variance, measured with squared daily log returns.

01 Forecast the baseline	02 Learn its errors	03 Evaluate both outputs
GARCH-t estimates next-day variance g using past returns.	LSTM predicts the signed error $e = y - g$ from lagged features.	Raw $h = g + \text{predicted } e$. Floored $h = \max(\text{raw } h, 1e-12)$.

The evidence at a glance



Paired stationary bootstrap: mean block length 21; 2,000 resamples. Intervals condition on the trained forecasts, not training-seed or model-selection uncertainty.

Raw MSE improves 5.45%, versus 5.48% when floored: the gain is not primarily a clipping effect.

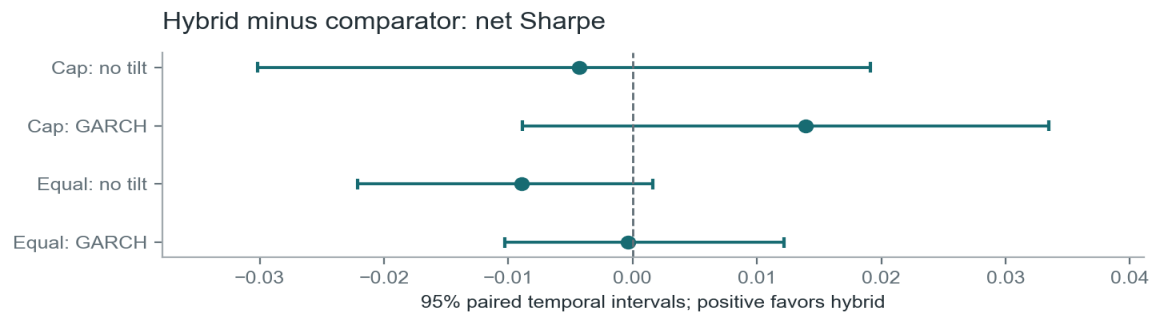
Validity: 60 raw forecasts are nonpositive. Floored QLIKE is 457,801, versus -8.4613 for GARCH; lower is better. Lower MSE alone is not a deployable-model result.

Does the LSTM Add Trading Value?

Weight = base weight x clip(trailing volatility / forecast, 0.5, 1.5), normalized. Compare the hybrid with both an untilted allocation and a GARCH-only tilt.

Monthly, 10 bps	Net Sharpe	Max drawdown
Snapshot-cap base	1.119	-33.25%
Snapshot-cap + GARCH	1.101	-33.15%
Snapshot-cap + GARCH/LSTM	1.115	-32.48%
Equal-weight base	0.738	-38.34%
Equal-weight + GARCH	0.729	-37.79%
Equal-weight + GARCH/LSTM	0.729	-38.15%

2,390 common return dates, 2015-07-06 to 2024-12-31. Next-close execution; 10 bps per dollar bought/sold; drifting holdings and self-financing costs. Sharpe assumes daily risk-free return 0.0001.



Snapshot-cap hybrid minus matched GARCH Sharpe: +0.0139; 95% interval [-0.0089, +0.0335]. The conditional interval includes zero. Mean bootstrap block 21 days, 2,000 paired resamples.

Matched controls and implementation

Both tilts use the same stock pool and model-availability dates. Unavailable or nonpositive forecasts receive neutral tilts. Unrestricted GARCH and all predefined frequency/cost sensitivities remain in the aggregate results.

What this does not establish

Forecast robustness failed: 475 nonpositive outputs and AMCR-driven extreme errors. Capping portfolio tilts does not repair the underlying forecast model.

Later constituents and repeated later market caps create survivorship and look-ahead bias. Equal weights do not fix the universe. These conditional comparisons are not validated alpha, achievable-return estimates or a causal proof of LSTM value.

Full report (8 pages) | Code, all sensitivities and source hashes

Source: reports/stock_retraining_results.json. Raw vendor data and per-stock model artifacts remain local. Based on collaborative research; report prepared by Eddie Liu.